

Maximum Likelihood Estimation

Bayesian Learning

Maximum Likelihood Estimation

- $D_1, D_2, \dots, D_c \rightarrow D_j$ is set of examples in class ω_j
- Class conditional probability distribution is $p(X|\omega_j)$
- Assume, $p(X|\omega_j)$ has a known parametric form, e.g., $p(X|\omega_j) \sim \mathcal{N}(\mu_j, \Sigma_j)$
 - The parameters are $\theta_j = \begin{bmatrix} \mu_j \\ \Sigma_j \end{bmatrix}$ So, $p(X|\omega_j)$ can also be written as $p(X|\omega_j, \theta_j)$
- **Objective:** Estimate θ_j from training samples in D_j for all j .
- Also assume, Samples in D_i does not provide any information of θ_j ; *if* $i \neq j$
- So
 - θ_1 is estimated from D_1 only
 - θ_2 is estimated from D_2 only
 - ...
 - θ_c is estimated from D_c only

} c no. of independent problems

So we can solve a general problem
Estimate θ that best describes
 $p(X|\theta)$ for all examples X in D

Distribution	Parameters
Bernoulli(p)	$\theta = p$
Poisson(λ)	$\theta = \lambda$
Uniform(a, b)	$\theta = (a, b)$
Normal(μ, σ^2)	$\theta = (\mu, \sigma^2)$
$Y = mX + b$	$\theta = (m, b)$

Maximum Likelihood Estimation

- Estimate θ that best describes $p(X|\theta)$ for all examples X in D
- **Or** Estimate θ that best describes $p(D|\theta)$
- Suppose D contains n examples: $\{X_1, X_2, \dots, X_n\}$ and examples are independent

Likelihood of θ
for the set D

$$p(D|\theta) = \prod_{k=1}^n p(X_k|\theta)$$

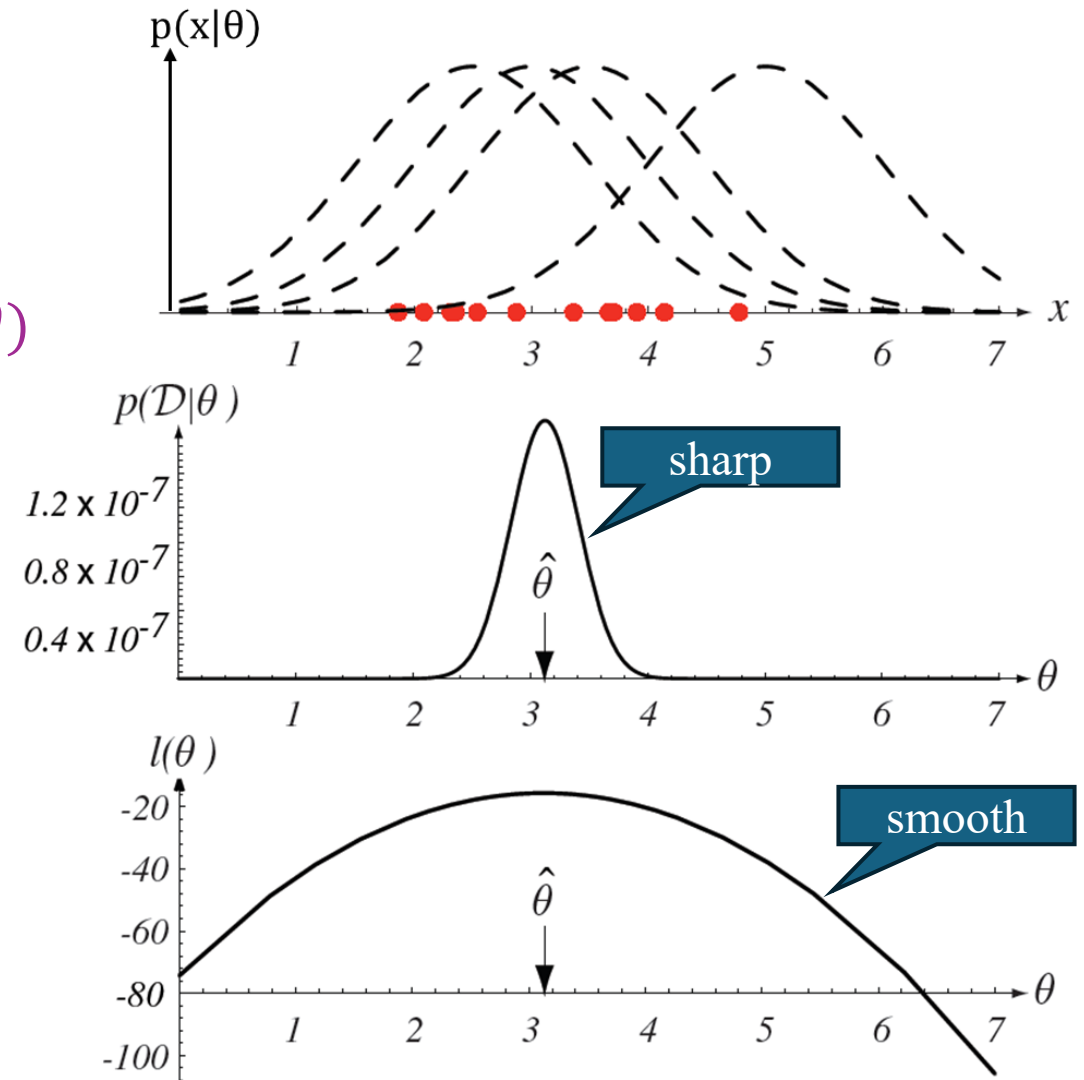
- **Objective:** Estimate $\hat{\theta}$ that maximizes $p(D|\theta)$ → *Maximum Likelihood estimate of θ*

Maximum Likelihood Estimation

Simple example of univariate case to estimate μ with known σ^2

- Likelihood: $p(D|\theta) = \prod_{k=1}^n p(X_k|\theta)$
- Log-likelihood: $\ln p(D|\theta) = \ell(\theta) = \sum_{k=1}^n \ln p(X_k|\theta)$
- We can also maximize $\ell(\theta)$

$$\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix} \quad \nabla_{\theta} = \begin{bmatrix} \frac{\partial}{\partial \theta_1} \\ \frac{\partial}{\partial \theta_2} \\ \vdots \\ \frac{\partial}{\partial \theta_p} \end{bmatrix} \quad \nabla_{\theta} \ell(\theta) = \begin{bmatrix} \frac{\partial \ell}{\partial \theta_1} \\ \frac{\partial \ell}{\partial \theta_2} \\ \vdots \\ \frac{\partial \ell}{\partial \theta_p} \end{bmatrix} = 0$$



Verify the MLE method from known cases

- $\ln p(D|\theta) = \ell(\theta) = \sum_{k=1}^n \ln p(X_k|\theta)$
- Given distribution $\mathcal{N}(\mu, \sigma^2)$, with σ^2 known
- MLE of μ ??

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right]$$

- $\ln p(X_k|\theta) = -\frac{1}{2} \ln 2\pi\sigma^2 - \frac{1}{2\sigma^2} (X_k - \mu)^2$

$$\nabla_{\theta} = \left[\frac{\partial \ell}{\partial \theta_1} \right] = \left[\frac{\partial \ell}{\partial \mu} \right] = \frac{\partial}{\partial \mu} \left(\sum_{k=1}^n \ln p(X_k|\theta) \right) = \sum_{k=1}^n \frac{\partial}{\partial \mu} \ln p(X_k|\theta) = 0$$

$$\Rightarrow \sum_{k=1}^n \frac{X_k - \mu}{\sigma^2} = 0$$

$$\Rightarrow \sum_{k=1}^n X_k - \sum_{k=1}^n \mu = 0$$

$$\Rightarrow \sum_{k=1}^n X_k = n\mu$$

$$\Rightarrow \mu = \frac{\sum_{k=1}^n X_k}{n}$$

Verify the MLE method from know cases

- $\ln p(D|\theta) = \ell(\theta) = \sum_{k=1}^n \ln p(X_k|\theta)$
- Known distribution $\mathcal{N}(\mu, \sigma^2)$, **both are unknown**
- MLE of μ and σ^2 ??

$$\bullet \theta = \begin{bmatrix} \theta_1 = \mu \\ \theta_2 = \sigma^2 \end{bmatrix}$$

$$\bullet \ln p(X_k|\theta) = -\frac{1}{2} \ln 2\pi\sigma^2 - \frac{1}{2\sigma^2} (X_k - \mu)^2$$

$$\bullet \nabla_{\theta} = \begin{bmatrix} \frac{\partial \ell}{\partial \theta_1} \\ \frac{\partial \ell}{\partial \theta_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial \ell}{\partial \mu} \\ \frac{\partial \ell}{\partial \sigma^2} \end{bmatrix} = 0$$

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right]$$

$$\frac{\partial \ell}{\partial \mu} = 0$$

$$\bullet \frac{\partial}{\partial \mu} (\sum_{k=1}^n \ln p(X_k|\theta)) = \sum_{k=1}^n \frac{\partial}{\partial \mu} \ln p(X_k|\theta) = 0$$

$$\bullet \Rightarrow \sum_{k=1}^n \frac{X_k - \mu}{\sigma^2} = 0$$

$$\bullet \Rightarrow \sum_{k=1}^n X_k - \sum_{k=1}^n \mu = 0$$

$$\bullet \Rightarrow \sum_{k=1}^n X_k = n\mu$$

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Verify the MLE method from know cases

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$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right]$$

$$\frac{\partial \ell}{\partial \sigma^2} = 0$$

$$\bullet \frac{\partial}{\partial \sigma^2} (\sum_{k=1}^n \ln p(X_k|\theta)) = \sum_{k=1}^n \frac{\partial}{\partial \sigma^2} \ln p(X_k|\theta) = 0$$

$$\bullet \Rightarrow \sum_{k=1}^n \frac{-1}{2\sigma^2} - \sum_{k=1}^n \frac{-1}{2(\sigma^2)^2} (X_k - \mu)^2 = 0$$

$$\bullet \Rightarrow \frac{1}{(\sigma^2)^2} \sum_{k=1}^n (X_k - \mu)^2 = \frac{n}{\sigma^2}$$

$$\bullet \sigma^2 = \frac{1}{n} \sum_{k=1}^n (X_k - \mu)^2$$

MLE for *pdf* parameter estimation

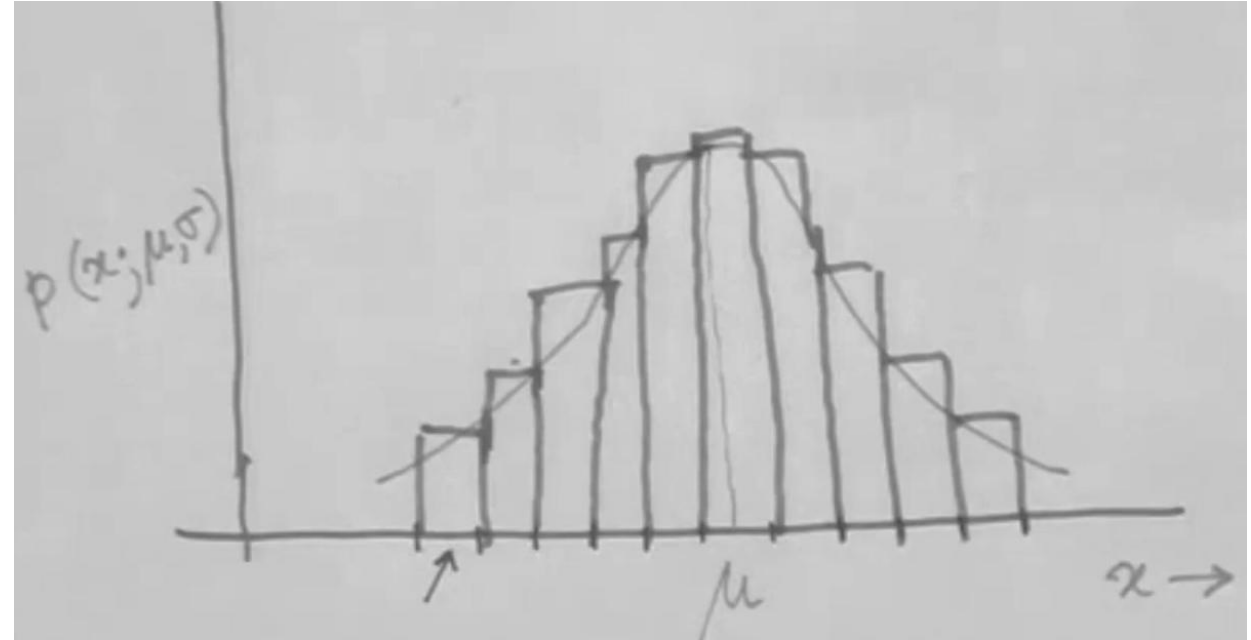
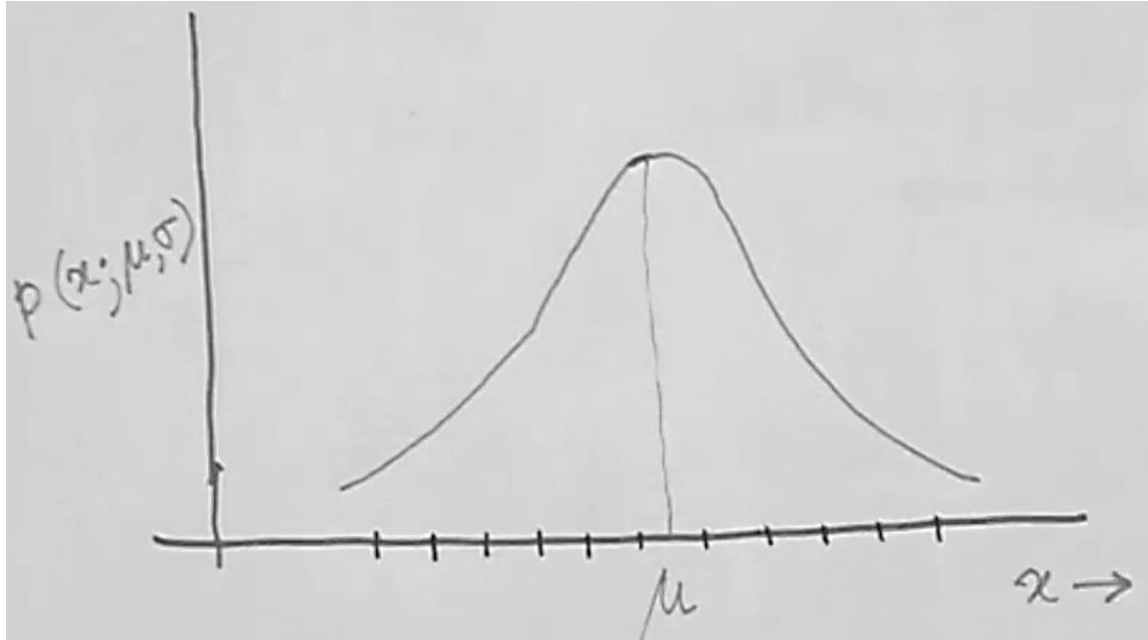
- MLE works only when the parametric form of the distribution is known.
- What if parametric form is un-known!!
- Can't we apply the Baye's classifier then?

$$g_i(\mathbf{x}) = p(\mathbf{x} | \omega_i) P(\omega_i)$$

$$g_i(\mathbf{x}) = \ln p(\mathbf{x} | \omega_i) + \ln P(\omega_i)$$

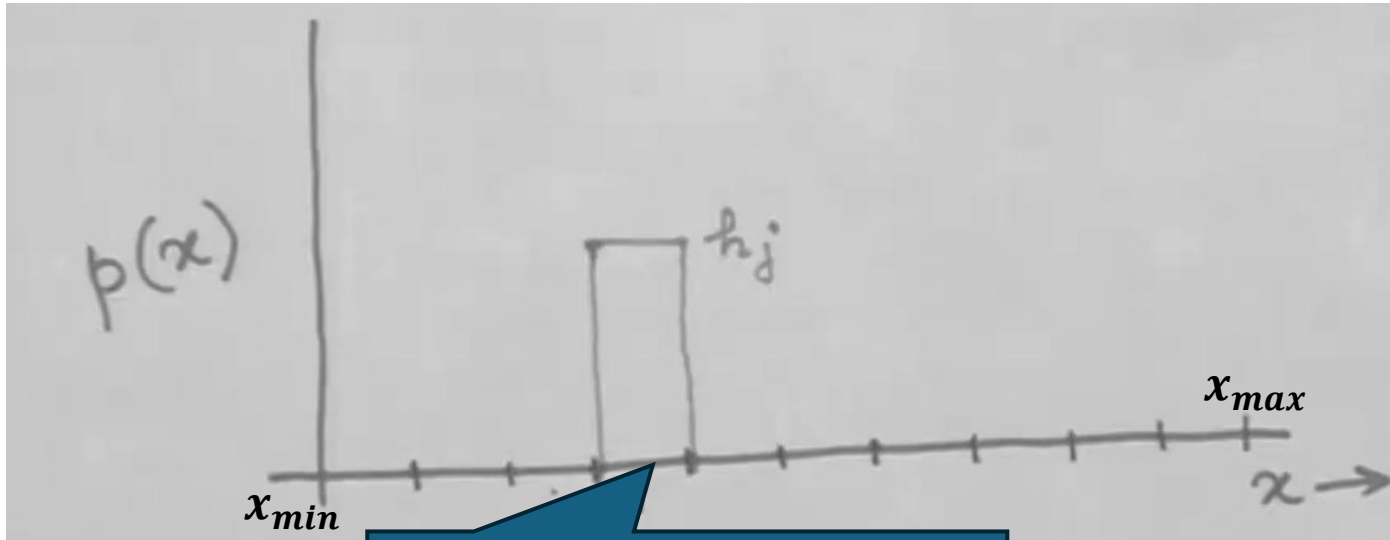
- Estimating $p(X|\omega_i)$ from the training samples in D_i [non-parametric pdf]
 - Histogram Technique

Idea: Piece-wise approximation of a gaussian



Motivation for Histogram Technique

Estimating $p(X|\omega_i)$ - Histogram Technique



Interval/bin/cell j of width w_j
(may be equal or un-equal w_j)

This is called *piece-wise approximation of the pdf*.

For an unknown X_1 , if it falls in bin_j then $P(X_1|\omega_1) = h_j$

N : Total no. of samples
 n_j : No of samples in bin_j
 h_j : height of bin_j

$$\text{Area of } bin_j = \frac{n_j}{N}$$

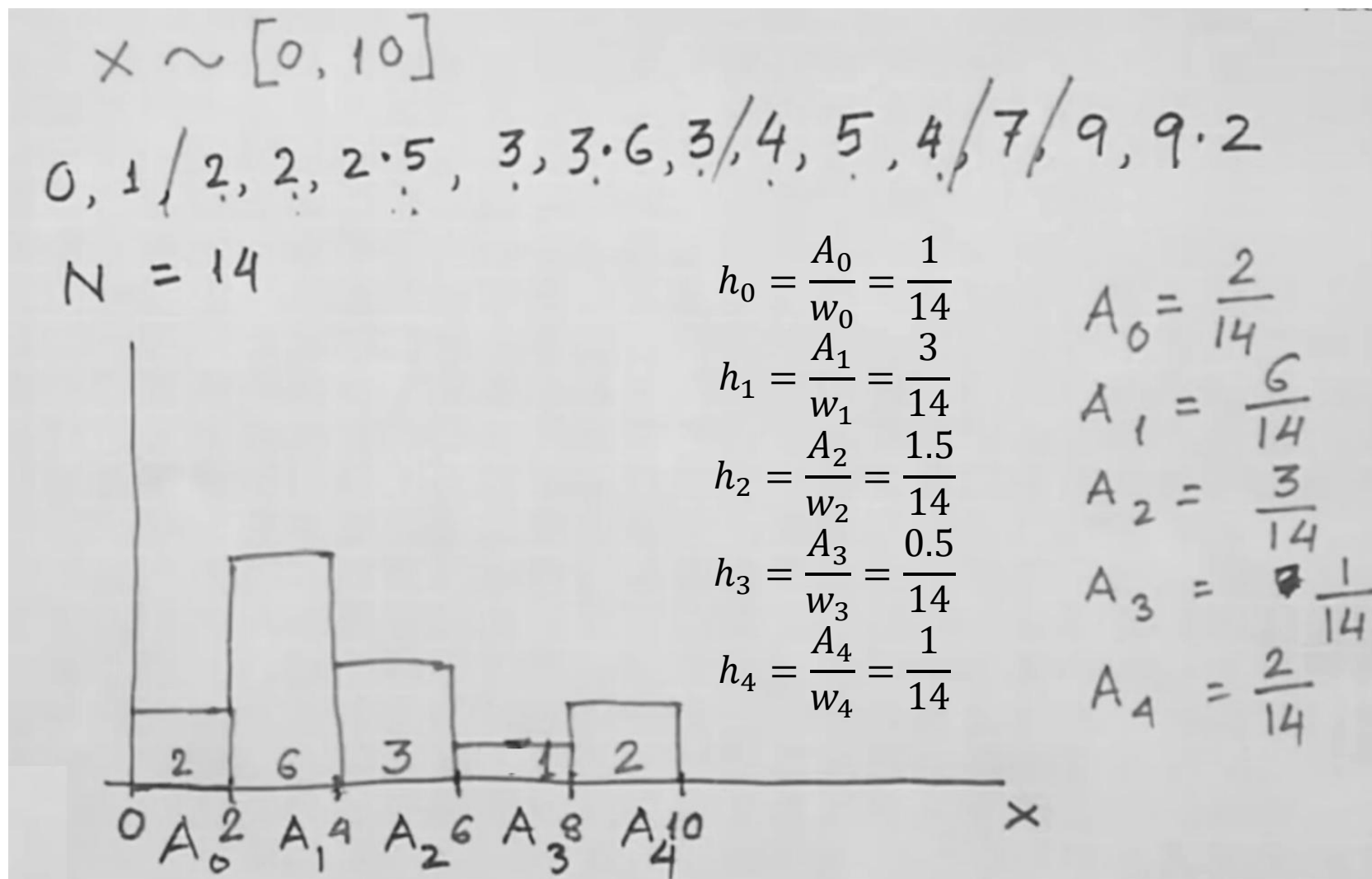
$$\sum_{\forall j} \text{Area}_j = 1$$

$$h_j = \frac{n_j}{N \cdot w_j}$$

Estimate of the prob.
density in the bin_j

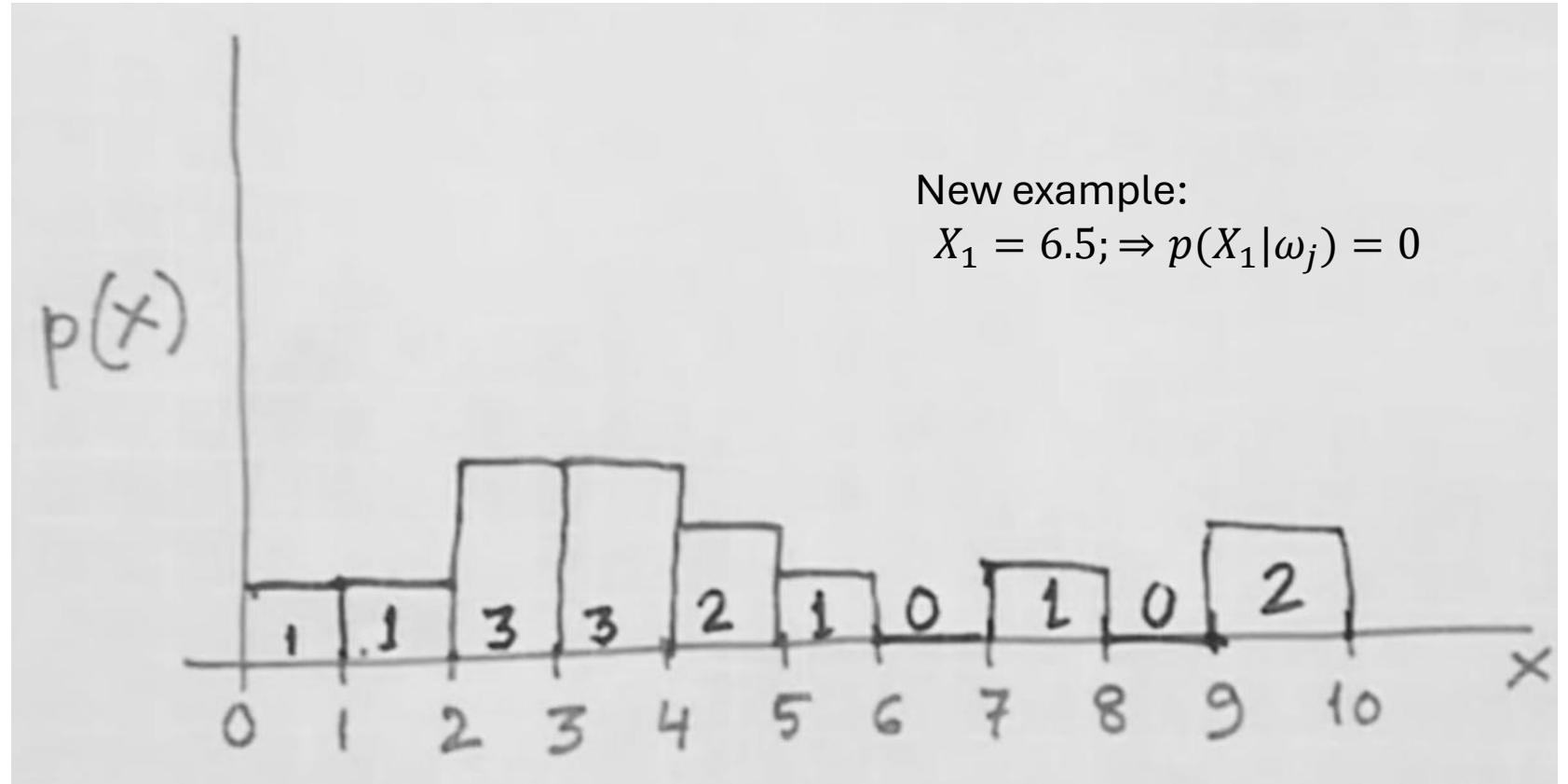
Estimating $p(X|\omega_i)$ - Histogram Technique

- $w_j = 2; \forall j$



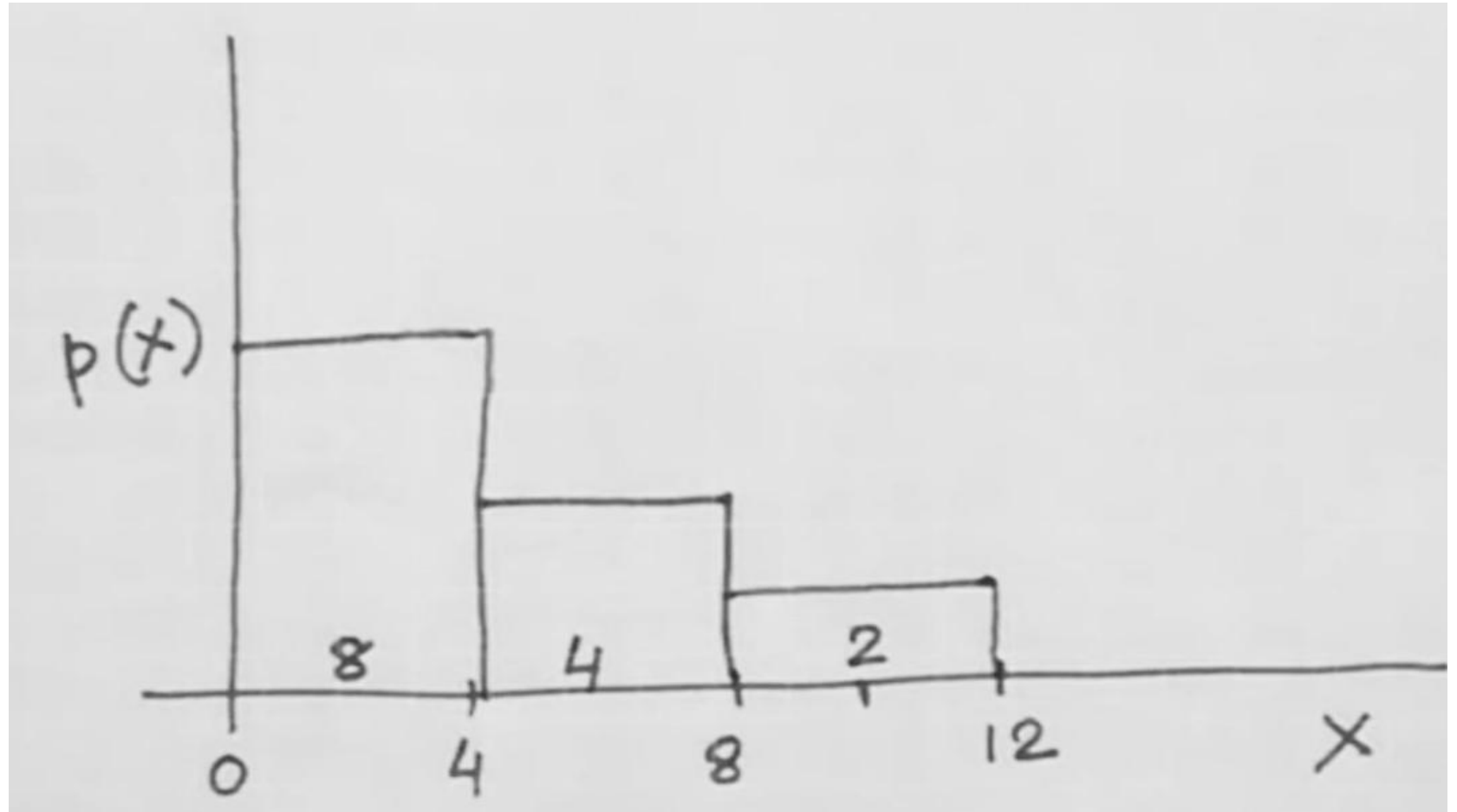
Estimating $p(X|\omega_i)$ - Histogram Technique

- $w_j = 1; \forall j$ (overfits)



Estimating $p(X|\omega_i)$ - Histogram Technique

- $w_j = 4; \forall j$ ((underfits))



Choosing bin-size

- There is no analytical tool to find the optimal bin-size
- However, a thumb-rule is available in the literature
- Bin-size = $\sqrt{N}$

Estimating $p(X|\omega_i)$ - Histogram Technique

$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ i. e. Two features

Volume under the *pdf surface* should be 1

$\frac{n_{ij}}{N}$: Volume of the pdf surface under the **bin**_{*i,j*} = *vol*_{*i,j*}

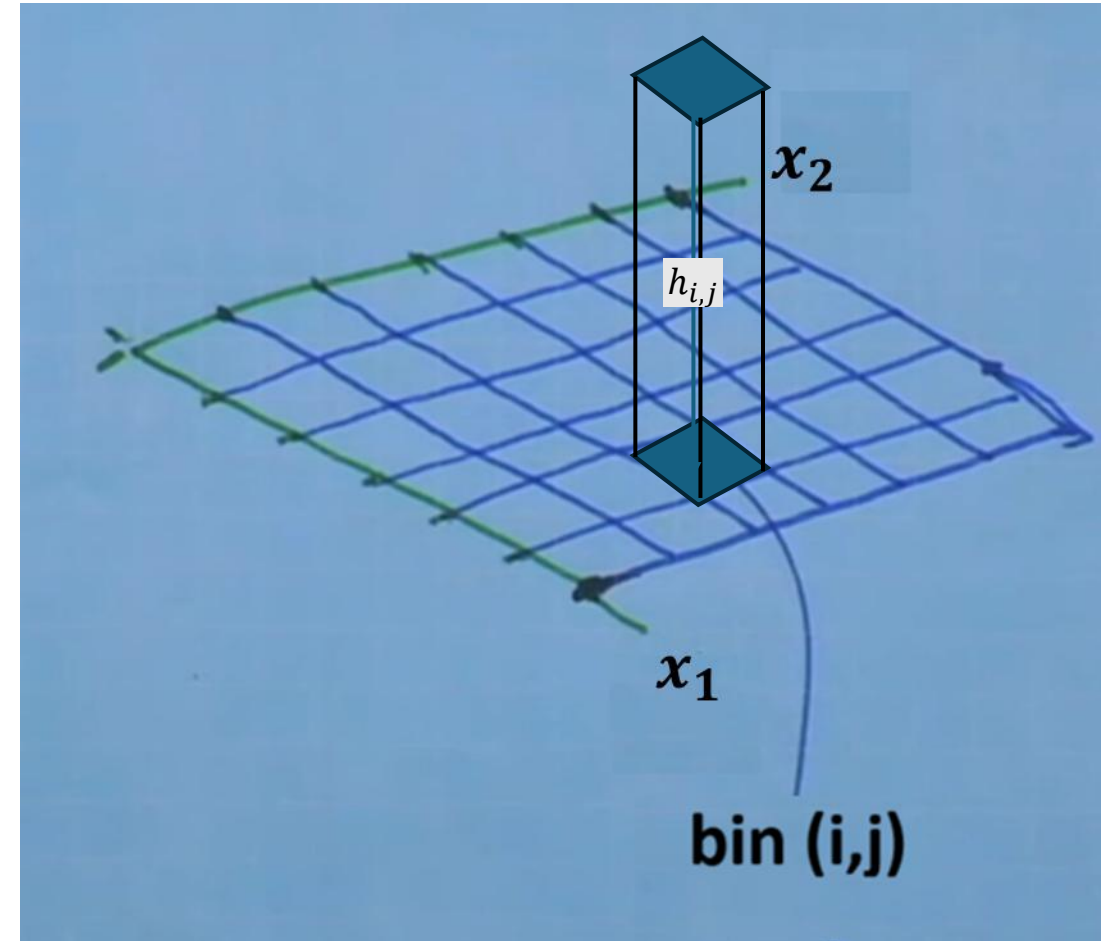
$$\sum_{\forall i,j} vol_{i,j} = \text{volume under pdf surface} = 1$$

$$h_{i,j} = \frac{vol_{i,j}}{area_{i,j}} = \frac{vol_{i,j}}{w_i \times w_j}$$

Given a new example $X_{new} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 8.4 \\ 4.5 \end{bmatrix}$

$$p(X_{new}|\omega_j) = h_{i,j}$$

[Assuming $w_i = w_j = 2$, and index starts from 0]



Estimating $p(X|\omega_i)$ - Histogram Technique

